

Figure 3: Kivy architecture [Source - Official Kivy Documentation from Kivy Docs, URL - kivy.org]

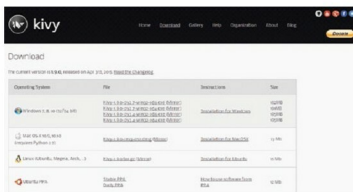


Figure 4: Timer clock app in Kivy

Architecture of Kivy

Kivy has a number of building blocks that work together for the development and execution of the mobile apps.

Creating a basic app

Use the basic app in Kivy to print a message on the mobile device. This is code that can be implemented by beginners:

```
from kivy.app import App
from kivy.uix.button import Button
class GoodMorning(App):
    def build(self):
        return Button(text='Good Morning to All of You')
GoodMorning().run()
```

Creating app code using Python Kivy - accessing the Web camera

This example demonstrates using the camera of the mobile device. The code will generate the window with the button 'Play' so that the camera can be moved to either the ON or OFF state.

```
from kivy.app import App
```

```
from kivy.lang import Builder
kv = '''
BoxLayout:
    orientation: 'vertical'
    Camera:
        id: camera
        resolution: (640, 480)
        play: False
    ToggleButton:
        text: 'Play'
        on_press: camera.play = not camera.play
        size_hint_y: None
        height: '48dp'
class MobileCamera(App):
    def build(self):
        return Builder.load_string(kv)
MobileCamera().run()
```

Creating code for a weather app

In this example, the code makes use of inheritance to create the subclass of WeatherApp.

```
from kivy.app import App
class MyWeatherAnalysisApp(App):
    pass
if __name__ == '__main__':
    MyWeatherAnalysisApp().run()
Creating a timer clock in Kivy
```

In this code, a very simple timer clock app is created. This is similar to the clock that is in-built with Android, iOS and other mobile platforms.

```
from kivy.app import App
class MyClockTimerApp(App):
    pass
if __name__ == '__main__':
    MyClockTimerApp().run()
# File: clock.kv
BoxLayout:
    orientation: 'vertical'
    Label:
        text: '00:00:00'
```

Kivy Garden

Garden is the specialised project, where all the add-ons and modules of Kivy can be centralised on a common point (<http://kivy-garden.github.io/>). Garden packages of Kivy are on the centralised GitHub repository.

Kivy Garden can be installed using the following command:

```
$ pip install kivy-garden
```